



South Carolina

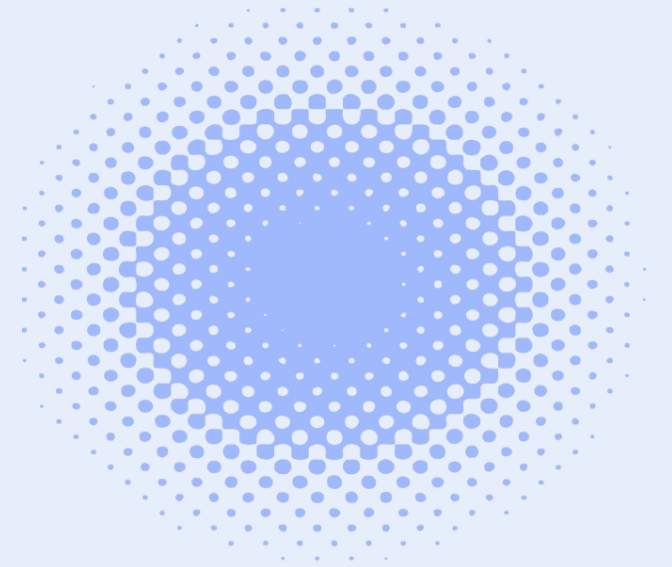
SCHEDULING ARCHITECTURE

Weighted Visit Assignment

How the system decides who goes on each family visit

Three layers, five weighted terms, one auditable number

Prepared for the lab team and PI • August 2026



Every assignment answers three questions

1



Who can go?

Some people are ruled out before anyone compares them. An open calendar slot, the right certification, no conflict with the family, and a way to get there.

2



Who fits best?

Fair rotation, family history, drive time, and current workload all matter at once. A number lets us weigh them together instead of one at a time.

3

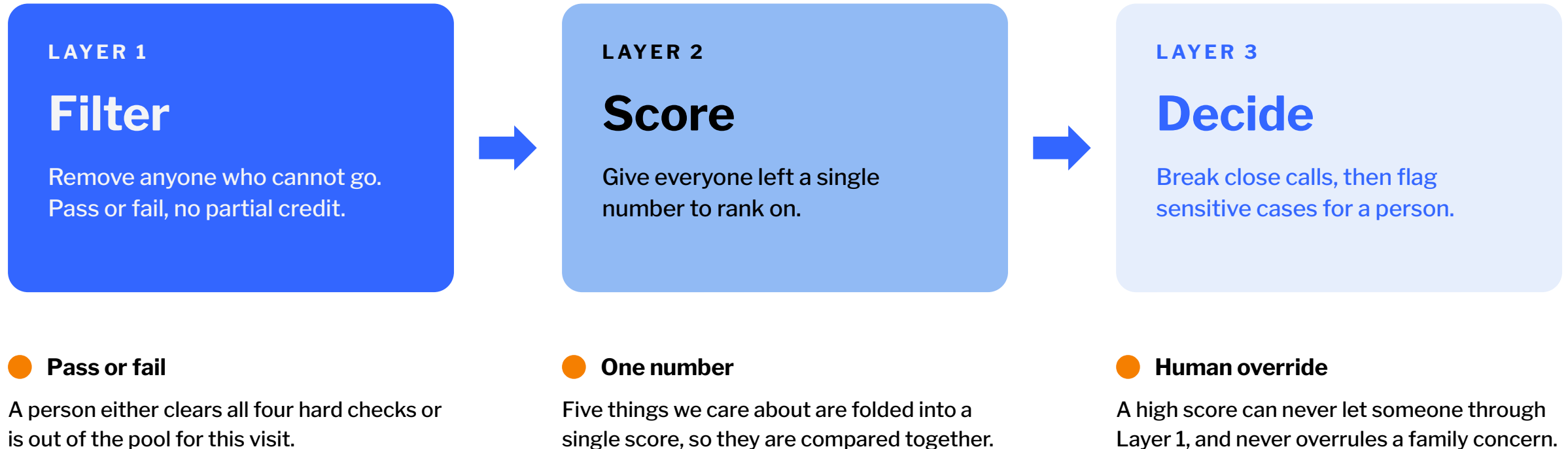


Who decides?

Close calls and sensitive family situations stop the automatic pick and go to the coordinator or the PI.

The three questions map onto three layers. A layer is never skipped, and a later layer never undoes an earlier one.

The system runs in three fixed layers



What each symbol stands for

$$i \in \mathcal{C}$$

One person who could take the visit

$$v \in \mathcal{V}$$

One visit request that needs filling

$$\Delta t_i$$

Days since that person's last assignment

$$n_{i,f}$$

Visits that person has already done with this family

$$\sigma_f \in \{-1, +1\}$$

Family flag: minus one wants a new face, plus one wants the same face

$$d_{i,v}$$

Round trip drive distance in miles

$$\mathcal{E}_v$$

The pool of people who cleared Layer 1

$$s_{i,v}$$

The final score for that person on that visit

$$\epsilon$$

How close two scores have to be to count as a tie

$$\gamma$$

The small bonus for having done the previous checkpoint

Four checks, and all four must pass

A

Open slot

The person has time free inside the requested date window.

E

No conflict

The person is not on this family's do not send list.

R

Certified

The person holds the credential this visit type requires.

G

Can get there

Transport works, including the van when the visit needs it.

$$\text{Eligible}(i, v) = A_{i,v} \wedge \neg E_{i,f} \wedge R_{i,v}^{\text{cert}} \wedge G_{i,v}$$

The wedge means and. The hooked bar means not. All four have to be true at the same time.
R carries a cert label here so it is not confused with the recency score in Layer 2.

If nobody clears all four, the visit goes straight to the coordinator for manual rescheduling. The score is never used to rescue an ineligible person.

A real case: two twin visits, one pool

Person	Open slot	No conflict	Certified	Transport	Result
Staff 1	Yes	On the list	Yes	Yes	Out
Staff 2	Yes	Clear	Yes	Yes	Stays in
Staff 3	No	Clear	Yes	Yes	Out

What happened

One person was on the family's exclusion list. One had no open slot. That left a single eligible person, who took both twin visits.

Why it matters

The pool collapsed to one before any scoring ran. Layer 1 did all the work, and the outcome matched what the team decided by hand.

Names are replaced with labels. The pattern is what matters, not who was involved.

One score, built from five parts

$$S_{i,v} = w_1 R_{i,v} + w_2 F_{i,f} + w_3 D_{i,v} + w_4 L_i + w_5 C_{i,f}$$

$w_1 = 0.30$

Recency

Longer since the last visit scores higher

$w_2 = 0.25$

Family history

Past visits with this family, read the way the family prefers

$w_3 = 0.20$

Travel

Shorter round trip scores higher

$w_4 = 0.15$

Workload

Fewer visits this period scores higher

$w_5 = 0.10$

Continuity

Small bonus for doing the previous checkpoint

$$\sum_{k=1}^5 w_k = 1, \quad w_k \geq 0$$

Weights add to one and every part is scaled to a common range, so no single part can dominate through its units.

$$\hat{i}_v = \arg \max_{i \in \mathcal{E}_v} S_{i,v}$$

Pick the eligible person with the highest score. Weights live in a settings table, so the PI can retune the policy without touching code.

Fair rotation and family history

1 Recency

weight 0.30

$$R_{i,v} = \frac{\min(\Delta t_i, T_{\text{cap}})}{T_{\text{cap}}}$$

Counts days since that person's last assignment, capped at a ceiling such as 90 days.

Someone who worked yesterday scores near zero. Someone free for three months scores one. This is what stops the same person being picked over and over.

2 Family history

weight 0.25

$$F_{i,f} = \sigma_f \cdot \frac{1}{1 + n_{i,f}}$$

Counts past visits with this family, then flips sign based on what the family prefers.

Set the flag to plus one and a familiar person scores higher. Set it to minus one and a fresh person scores higher. The family decides, not the model.

Travel, workload, and continuity

3 Travel

weight 0.20

$$D_{i,v} = 1 - \frac{d_{i,v}}{d_{\max}}$$

Compares this drive to the longest drive in the eligible pool.

The shortest trip scores one. The longest scores zero. Everyone else sits in between.

4 Workload

weight 0.15

$$L_i = 1 - \frac{V_i^{(\text{period})}}{V_{\max}^{(\text{period})}}$$

Compares visits done this period to the busiest person on the team.

The busiest person scores zero, so the load spreads out over a review period rather than a single week.

5 Continuity

weight 0.10

$$C_{i,f} = \begin{cases} \gamma, & \text{last checkpoint} \\ 0, & \text{otherwise} \end{cases}$$

A small fixed bonus, about 0.05, for doing the last checkpoint.

Deliberately small. It nudges toward a familiar face without ever overturning a fair rotation on its own.

Two eligible people, side by side

What we measure	Candidate A	Candidate B	Who it favours
Days since last assignment	5	20	B
Recency score	0.056	0.222	B
Past visits with this family	1	0	A
Family history score (flag is plus one)	0.500	1.000	B
Round trip miles	8	22	A
Travel score	0.636	0.000	A
Visits already this period	6	3	B
Workload score	0.000	0.500	B
Did the previous checkpoint	Yes	No	A
Continuity bonus	0.050	0.000	A

Rows in blue are the five part scores. Everything above them is the raw measurement they come from.

The weights in force

$(w_1, w_2, w_3, w_4, w_5) = (0.30, 0.25, 0.20, 0.15, 0.10)$

Read left to right: recency, family history, travel, workload, continuity.

Split decision

A wins on travel and continuity. B wins on recency, family history, and workload. Nobody wins outright, which is exactly the case a single number is built for.

Illustration only, not a real assignment.

Multiply, add, compare

CANDIDATE A

$$S_A = 0.30(0.056) + 0.25(0.500) + 0.20(0.636) + 0.15(0.000) + 0.10(0.050) = 0.274$$

CANDIDATE B

$$S_B = 0.30(0.222) + 0.25(1.000) + 0.20(0.000) + 0.15(0.500) + 0.10(0.000) = 0.392$$

SELECTED

● What this shows

Candidate B takes it by 0.118. A longer gap since the last visit, a lighter caseload, and a clean slate with this family together outweigh B's much longer drive. Continuity alone was never going to carry Candidate A, and that is the point of scoring all five parts at once instead of ranking them one after another.

What happens when two scores are almost equal

STEP ONE • DETECT

$$|S_{i,v} - S_{j,v}| < \varepsilon$$

WHAT IT MEANS

If the top two scores sit inside the tolerance, the ranking is not meaningful and the model stops pretending it is.

STEP TWO • RESOLVE ON FAMILY HISTORY

$$\text{Winner} = \begin{cases} \arg \min_i n_{i,f}, & \sigma_f = -1 \quad (\text{new face preferred}) \\ \arg \max_i n_{i,f}, & \sigma_f = +1 \quad (\text{same face preferred}) \end{cases}$$

Family wants a new face

The person with the fewest past visits wins the tie.

Family wants the same face

The person with the most past visits wins the tie.

Close but not tied

Still assign, but flag it so a coordinator can look before it is confirmed.

When a person decides instead of the score

YES

Does the family have a dual relationship flag?

Remove that person from the pool. No score is computed for them at all.

YES

Is there a PI hold on the family record?

Pause the automatic pick and route the visit to the PI for a manual decision.

GO

Neither flag is set

Score the eligible pool, pick the top person, and confirm automatically.

These checks run before scoring, never after. A family concern is a stop sign, not a penalty term.

Eight tables feed the whole model

Table	Key fields	Feeds
availability	candidate id, date start, date end	Layer 1 open slot check
family exclusions	family id, excluded candidate id, reason	Layer 1 conflict check
certifications	candidate id, visit type, certified	Layer 1 credential check
visit history	visit id, family id, candidate id, date, visit type	Recency, family history, continuity
distance matrix	candidate id, family id, distance miles	Travel score
workload period	candidate id, period, visit count	Workload score
family continuity flag	family id, sigma, preferred candidate id	Family preference and overrides
scoring weights	w1 to w5, effective date	Versioned policy settings

Only the last two tables are new policy. The first six are records the lab already keeps, which is why this can be built without changing how anyone works day to day.

THE WHOLE RULE, IN ONE LINE

No feasible pool, no assignment.
No assignment without a score you can read.

$$\text{Assignment}(v) = \begin{cases} \text{send to coordinator,} & \mathcal{E}_v = \emptyset \\ \arg \max_{I \in \mathcal{E}_v} S_{I,v}, & \text{otherwise} \end{cases}$$

Explainable

Every pick reduces to five numbers and five weights that anyone can check by hand.

Tunable

The PI changes the weights in a settings table.
No code release, and old picks stay reproducible.

Honest

Close calls and family concerns come back to a person, by design rather than by accident.